P-W-158E
April 30, 1977
SUPERSEDING
Int. Fed. Spec. P-W-00158D
October 23, 1964 and
Fed. Spec. P-W-158C
January 2, 1964

FEDERAL SPECIFICATION

WAX, GENERAL PURPOSE, SOLVENT TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types and two classes of general purpose solvent waxes (see 6.1).

1.2 Classification.

1.2.1 Types. The waxes covered by this specification shall be of the following types and classes, as specified (see 6.2).

Type I - Liquid
  Class 1 - Liquid without silica
  Class 2 - Liquid with silica

Type II - Paste

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-P-704 - Pails, Metal; (Shipping, Steel, 1 through 12 Gallons).

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 7930

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Hazardous Substances Act Regulations:

16 CFR 1500 - Hazardous Substances and Articles: Administration and Enforcement Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Designations:

D 56 - Test for Flash Point by Tag Closed Tester.
D 217 - Test for Cone Penetration of Lubricating Grease.
D 1386 - Test for Acid Number (Empirical) of Natural Waxes.
D 1387 - Test for Saponification Number (Empirical) of Natural Waxes.
D 1542 - Test for Rosin in Varnishes, Qualitative.
D 2834 - Test for Nonvolatile Matter (Total Solids) in Water-Emulsion Floor Polishes, Solvent-Based Floor Polishes, and Polymer-Emulsion Floor Polishes.
E 28 - Test for Softening Point by Ring-and-Ball Apparatus.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Types I and II wax shall consist of natural or synthetic waxes or a mixture thereof dissolved or dispersed in a volatile organic solvent consisting of turpentine or volatile petroleum distillates or a mixture thereof, suitable for application at 18° to 30°C (64.4° to 86°F).

3.1.1 Type I. Type I material shall be a homogeneous free-flowing liquid.

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3.1.2 Type II. Type II material shall be a soft, homogeneous paste, having a cone penetration range of 60 to 140 at 30° ± 0.5°C (86° ± 1.0°F) when tested as specified in 4.4.1.

3.2 Chemical properties. When tested according to 4.4, types I and II material shall conform to the requirements set forth in table I.

TABLE I. Chemical properties 1/

Characteristic	Type I				Type II	
	Class 1		Class 2			
	Min.	Max.	Min.	Max.	Min.	Max.
Softening point	77°C	100°C	77°C	100°C	77°C	100°C
Acid number	0	25	0	25	0	25
Saponification number	20	100	20	100	20	100
Ash content, percent	0.0	0.5	1.55	1.70	0.0	0.0
Rosin	None	None	None	None	None	None
Silica, percent	--	--	1.55	1.65	--	--
Nonvolatile content, percent	11.0	--	9.75	--	20.0	--
Flash point	60°C	--	60°C	--	60°C	--

1/ The softening point, acid number, and saponification number shall be run on the nonvolatile portion; and on the class 2 wax shall be determined after the removal of the silica from the nonvolatile portion.

3.3 Physical properties.

3.3.1 Abrasive. Types I and II material shall be free of grit or abrasive matter when tested as specified in 4.4.9.

3.3.2 Stability. Types I and II material shall show no signs of separation or (in type I) settling when tested as specified in 4.4.10.

3.3.3 Film properties. Types I and II waxes shall dry to a semi-transparent, non-tacky film within 45 minutes. The film shall be free from whiteness or cloudiness, when tested as specified in 4.4.11.

3.3.4 Drying time. When tested as specified in 4.4.12, type I wax shall be dry to the touch at the end of 15 minutes.

3.3.5 Performance test. Type I and type II material shall meet the performance test specified in 4.4.13.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection. Sampling for inspection and tests shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated.

4.3 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

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TABLE II. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper, or missing.

4.3.1 Examination of closure, waterproofing, and banding for containers. When containers are required to comply with PPP-C-96, examination for defects in closure, waterproofing, and banding shall be in accordance with the appendix to that specification.

4.4 Testing the end item. The end item shall be tested for the characteristics specified in table I according to the test methods specified in table III. The tests shall be performed in duplicate on a composite sample consisting of five 1-pint samples from five separate containers of type 1 material or five 8-ounce samples of type II material similarly composited.

TABLE III. Instructions for testing end item (composite)

Characteristic	Specification reference: Requirement	Specification reference: Test method	Results reported as: Pass or fail 1/	Results reported as: Numerically to nearest
Consistency (type II)	3.1.2	4.4.1		1.0 unit
Nonvolatile matter	Table I	4.4.2		0.1 percent
Silica content	Table I	4.4.3		0.5 percent
Softening point of nonvolatile matter	Table I	4.4.4		1.0°C
Acid value of nonvolatile matter	Table I	4.4.5		1.0 unit
Saponification value of nonvolatile matter	Table I	4.4.6		1.0 unit
Ash content	Table I	4.4.3		0.05 percent
Rosin	Table I	4.4.7	X	-
Flash point of solvent	Table I	4.4.8		1.0°C
Abrasives	3.3.1	4.4.9	X	-
Stability	3.3.2	4.4.10	X	
Film properties	3.3.3	4.4.11	X	
Drying time	3.3.4	4.4.12	X	
Performance test	3.3.6	4.4.13	X	

1/ If failure is indicated report either description of failure or numerical point of failure as applicable.

4.4.1 Consistency. The cone penetration of type II material shall be determined according to ASTM D 217.

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4.4.2 Nonvolatile matter. Nonvolatile matter (total solids) shall be determined in accordance with ASTM D 2834.

4.4.3 Ash and silica content. Weigh a 10-gram sample into a pre-ignited and tared platinum crucible. Add sufficient ashless cotton to absorb the sample and ignite slowly at first over a Meker burner. Place in a muffle furnace at 800°C for 30 minutes. Cool, desiccate, and weigh. This is the weight of ash. Moisten the residue with 1 to 2 ml. of water, add 4 to 5 drops of sulfuric acid (1 part acid-2 parts water) and approximately 5 ml. of hydrofluoric acid. Evaporate to SO_3 fumes and finally ignite, cool, desiccate, and reweigh.

$$\text{Percent ash} = \frac{\text{Weight of ash} \times 100}{\text{Sample weight}}$$

$$\text{Percent silica} = \frac{\text{Loss in weight (weight of ash - final weight)} \times 100}{\text{Sample weight}}$$

4.4.4 Softening point. The dried residue shall be obtained in accordance with 4.4.2, except that the sample shall be sufficient in quantity to result in an ample residue for determining the softening point in accordance with ASTM E 28.

4.4.5 Acid value. The acid value shall be determined in accordance with ASTM D 1386.

4.4.6 Saponification value. The saponification value shall be determined in accordance with ASTM D 1387.

4.4.7 Rosin. The wax shall be tested for the presence of rosin in accordance with the Liebermann-Storch Test in ASTM D 1542.

4.4.8 Flash point of the volatile organic solvent. Place 80 to 100 grams of the sample wax in a distilling flask and distill off the solvent. The flash point of the distillate shall be determined in accordance with ASTM D 56.

4.4.9 Abrasive properties. Spread a thin film of wax over a microscope slide. Place another microscope slide over the film of polish. Press the two slides together using a slight pressure and at the same time sliding one glass slide over the other. The movement between the glass slides shall be smooth without any indication of the presence of abrasive.

4.4.10 Stability. Types I and II material shall be cycled between 30°C (86°F) and -10°C (14°F) three times. One cycle shall consist of 6 hours at 30°C and 6 hours at -10°C. The degree of settling of suspended matter in type I material shall be noted. There shall be no separation of solvent from the type II material.

4.4.11 Properties of the film. Take a clear glass plate of a thickness not more than 6.3 mm (1/4 inch). Using a small sponge or cloth spread a thin layer of wax uniformly over 257 sq cm (40 sq in) of the surface of the plate. Allow to dry in a horizontal position for 45 minutes at a temperature of 20° to 25°C (68° to 77°F) and a relative humidity of 50 to 60 percent. Polish the plate with a dry cloth. The film shall be free from tackiness and shall show no signs of whiteness. When the film is viewed through the underside of the glass plate and by bright daylight transmitted through it, the film shall be from semi-transparent to clear.

4.4.12 Drying time (type I). Under conditions of 20° to 25°C (68°F to 77°F), and a relative humidity of 50 to 60 percent, apply 5 grams of the wax with a cheesecloth pad (4 ply 3 x 3 inch) over 1084 sq cm (168 sq in) of a well sealed, glass lacquered, wood panel. Observe for compliance with 3.3.4.

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4.4.13 Performance.

4.4.13.1 Apparatus. Gardner Washability Machine 1/, or equal, Gardner Roller 2/, or equal, CSMA standard soil 3/ and white homogenous vinyl tile.

1/ Available from Gardner Laboratory Inc., 5521 Landy Lane, Bethesda, MD Catalog No. WG-2000.
2/ Catalog No. WG-2000-R.
3/ Standard soil is available from the Chemical Specialities Manufacturer's Association, 1001 Connecticut Ave, NW, Washington, DC 20036.

4.4.13.2 Procedure. Cut the tile to fit the washability machine and clean the cut tiles thoroughly. Apply the sample with cotton gauze or lamb's wool, allow to dry for 15 minutes, and buff lightly with lamb's wool; then apply a second coat in the same manner. Age the tiles for 48 hours before use. Spread 2.00 grams of standard soil as evenly as possible over portion of the tiles where the roller will track. Run the machine 300 cycles (600 passes), brushing the soil back on the track as necessary. At the end of the cycle, remove any loose soil by wiping the surface with a facial tissue. Remove the soiled tile and apply a third coat of type I or type II material, rubbing the tile lightly with the cotton gauze or lamb's wool applicator. The tiles should appear clean and free from soil and when dry should have a lustrous surface, free from tackiness, streaks, or smears.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or Commercial, as specified (6.2).

5.1.1 Level A.

5.1.1.1 One-gallon quantity. One gallon of type I material shall be furnished in a one-gallon oblong can conforming to PPP-C-96, type V, class 4, having a screw-cap closure with an inner seal and provided with a bridge type handle. The can shall be closed in accordance with the appendix to PPP-C-96.

5.1.1.2 One-pound and four-pound quantity. One pound or four pounds of type II material shall be packaged in a can conforming to PPP-C-96, type V, class 2. The can shall be closed in accordance with the appendix to PPP-C-96.

5.1.2 Commercial. The wax shall be preserved and packaged in accordance with normal commercial practice.

5.2 Packing. Packing shall be level A, or Commercial, as specified (6.2).

5.2.1 Level A.

5.2.1.1 One-gallon quantity. Six cans of type I material shall be packed in a box in accordance with the level A packing requirements of the appendix of PPP-C-96.

5.2.1.2 Five-gallon quantity. Five gallons of type I material shall be furnished in a pail conforming to PPP-P-704, type I, class 8, with a screw cap closure and an inner and outer seal. The pail shall have an interior and exterior protective coating. The color of the exterior coating shall match color 24087 of Fed. Std. No. 595.

5.2.1.3 One-pound and four-pound quantity. The cans, specified in 5.1.1.2, shall be packed in a box in accordance with the level A packing requirements of the appendix of PPP-C-96, except the four-pound can size shall be packed as specified for the five-pound can size.

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5.2.1.4 Thirty-pound quantity. Thirty pounds of type II material shall be furnished in a pail conforming to PPP-P-704, type II, class 3. The pail shall have an interior and an exterior protective coating. The color or the exterior coating shall match 24087 of Fed. Std. No. 595.

5.2.2 Commercial. The wax shall be packed in shipping containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.3 Marking.

5.3.1 Labeling. The container shall bear clear and concise directions for use and the phase "Do not use on asphalt, vinyl, or rubber tile floors." In addition, the statement "CAUTION: Combustible Mixture" shall appear on the label in a prominent position in not smaller than 8-point capital letters. Labeling data shall be in accordance with the Federal Hazardous Substances Labeling Act.

5.3.2 Civil agencies. In addition to labeling specified in 5.3.1, unit and shipping containers and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123.

5.3.3 Military requirements. In addition to labeling specified in 5.3.1, unit and shipping containers and unitized loads (when applicable) shall be marked in accordance with MIL-STD-129.

5.4 Unitization. (Applicable to full rail car and truck load shipments only.) The wax shall be unitized for shipment in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length and 45 inches in width.

6. NOTES

6.1 Intended use. Types I and II material are intended primarily for use on wooden or cork floors, but may be used on painted or sealed wood floors, terrazzo, ceramic, marble and tile. They are not recommended for use on asphalt, vinyl or rubber composition floor tiles.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of wax desired (see 1.2.1 and 6.1).
(c) Packaging and packing required (5.1 and 5.2).
(d) Marking required (5.3).

MILITARY INTEREST:

The Department of Defense waives coordination on future revisions or amendments of this specification until further notice.

User activity:

Army - GL

Preparing Activity

GSA-FSS

Civil Agency Coordinating Activities:

GSA-FSS, PBO
VA-DMS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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